

NODE 3 — SELF-SCAN EXTERNAL GEOMETRY MEASUREMENT DEVICE

A. Overview

In certain embodiments, a third node is configured as a self-scan external geometry measurement device configured to measure external geometry parameters including one or more of length, diameter, shape profile, and related dimensional features. The third node may be used in a guided session and may generate repeatable, privacy-preserving geometry metrics suitable for baseline establishment, longitudinal comparison, and cross-device correlation with one or more other nodes.

B. Form Factor and Scan Geometry

The third node may be implemented as a partially open ring having a C-shape geometry or as a fully enclosed ring. The device may be moved along an axis during a scan routine, and may include internal contact surfaces configured to maintain stable positioning while minimizing friction and discomfort. In certain embodiments, the device includes a handle, grip region, docking base, or alignment features to encourage consistent scan paths and repeatable measurements. In certain embodiments, the device includes internal reference points or geometric constraints configured to control scan angle, scan speed, and scan position. The device may include compliance features, low-friction liners, or discrete contact pads to improve repeatability.

C. Sensor Suite

The third node may include one or more sensors and sensing elements, including combinations of the following: 1. Optical flow sensing for length determination, including one or more optical flow sensors arranged to detect motion along the scan axis. Optical flow sensing may be implemented using mouse-style motion sensors or other optical motion sensors. 2. Diameter sensing using one or more time-of-flight sensors, lidar, and/or infrared sensors arranged to measure distance to a surface and derive diameter or width profiles. 3. Inertial sensing including a 6-axis IMU to measure motion state, orientation, scan angle, and to support artifact rejection and scan validation. 4. Anti-spoof sensing, optionally including optical pulse sensing and/or pressure/contact sensing to provide physiologic confirmation and/or valid-context confirmation. 5. Contact sensing, optionally including FSR or other pressure/contact sensors, capacitive contact sensors, or proximity sensors, to confirm contact state and scan validity. Sensors may be arranged as a single sensor set or multiple zones. In some embodiments, multiple optical flow sensors are spaced apart to improve robustness, confirm scan direction, and detect slippage. In some embodiments, multiple time-of-flight or lidar sensors are distributed to capture a diameter profile across multiple angles.

D. Scan Routine, Validation, and Artifact Rejection

The third node may operate in a session-based mode in which the user performs a guided scan routine. The computing system may validate scan quality using one or more of IMU-derived scan angle constraints, scan speed constraints, contact confirmation, optical coupling checks, and multi-sensor agreement. The system may compute a scan quality index and may reject segments exhibiting slippage, discontinuity, invalid orientation, poor contact, or inconsistent readings.

E. Derived Features and Metrics

The computing system may derive geometry features including length measures, diameter measures, circumference estimates, taper measures, segmental profiles, symmetry measures, and repeatability measures. Metrics may be presented as privacy-preserving outputs including indices, ranges, and profile curves without explicit anatomical imagery. Example outputs include: a length estimate and associated confidence indicator, a diameter profile curve comprising multiple points along a scan path, a shape stability or repeatability score based on repeated scans, and a scan confidence indicator based on sensor agreement and artifact rejection.

F. Cross-Device Correlation

Node 3 outputs may be correlated with longitudinal physiology outputs from other nodes to normalize and interpret patterns using geometry context and repeatable structural parameters. In certain embodiments, Node 3 outputs are correlated with baseline acquisition outputs and/or intraluminal geometry/compliance outputs to support cross-user correlation and pairing-context analysis using privacy-preserving metrics.

G. Variations and Alternatives

The third node may be implemented with different sensor combinations. Certain embodiments emphasize optical flow for length and time-of-flight for diameter; other embodiments emphasize lidar; other embodiments include additional anti-spoof sensors or contact confirmation sensors. The device may include a docking base for charging, alignment, or calibration. Processing may be performed locally or via a computing system. The node may store data locally and upload periodically or stream in real time.